

# Odd Discrete Tori in Trigonometric Chaos

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## Status

This packet gives a likely valid full answer to the extracted signal in Remark 2.4 of Cano-Marmol, Conde-Alonso, and Parcet [1]: an explicit orthonormal basis for the cocycle Hilbert space of the odd cyclic product  $\mathbb{Z}_{2m+1}^n$ , and the resulting balanced Fourier-truncation inequality. The result is a completion of a source-paper computation left to the interested reader, not a claim that a separate major conjecture is resolved.

## Source Signal

The source paper proves the even-cardinal discrete-torus case in Section 2.2 and then records the following odd-cardinal variant in Remark 2.4 on page 16.

since  $\sigma = \min_{k \in [n]} \psi(\{k\}) = 4 \min_{k \in [n]} \alpha_k$ . Thus taking  $\alpha_j = 1$  for all  $j$  is optimal.

**Remark 2.4.** It is natural to ask if the situation changes much when the cyclic groups under consideration have odd cardinal. The function  $\psi(g) = \sum_{j=1}^n |g_j|$  with  $|g_j| = \min\{g_j, 2m+1-g_j\}$  is a conditionally negative length on  $\mathbb{Z}_{2m+1}^n$ , and so there exists an associated cocycle induced by the Gromov form

$$\begin{aligned} \langle \delta_g, \delta_h \rangle &= \frac{1}{2} (\psi(g) + \psi(h) - \psi(g-h)) \\ &= \sum_{j=1}^n \min \left\{ g_j, 2m+1-h_j, \max \left\{ 0, m-h_j+g_j+\frac{1}{2} \right\} \right\}. \end{aligned}$$

It defines a cocycle Hilbert space  $\mathcal{H}_\psi$  with dimension  $2mn$ . Theorems [A](#) and [B](#) apply but calculating an explicit expression for the orthonormal basis of  $\mathcal{H}_\psi$  is more tedious and we shall leave it to the interested reader.

In words, the authors note that for  $\mathbb{Z}_{2m+1}^n$  the word length

$$\psi(g) = \sum_{j=1}^n |g_j|, \quad |a| = \min\{a, 2m+1-a\},$$

is conditionally negative, that the associated cocycle Hilbert space has dimension  $2mn$ , and that Theorems A and B apply; what is left open in the remark is an explicit orthonormal basis.

## Proof Intuition

For the even cycle, the source basis is made of oriented edges, but opposite edges cancel in pairs. In an odd cycle there is no antipodal pairing, so the edge basis is no longer orthogonal in the same elementary way. The clean replacement is to diagonalize the cycle distance by the nonzero Fourier modes of  $\mathbb{Z}_{2m+1}$ . The negative Fourier coefficients of the distance give the normalizing constants. Taking real and imaginary parts produces  $2m$  real orthonormal vectors for each coordinate. These coordinate Fourier planes are rotated by the cocycle action, so they give the same admissible coordinate splitting used by the source paper. The coordinate support derivative  $\partial_j \lambda(g) = \mathbf{1}_{\{g_j \neq 0\}} \lambda(g)$  then absorbs every cocycle derivative in the  $j$ th coordinate, which is exactly the distinguished derivative hypothesis of Theorem B in [1].

## Theorem

Let  $N = 2m + 1$  and write elements of  $\mathbb{Z}_N$  as  $0, 1, \dots, N - 1$ , with arithmetic modulo  $N$ . Put

$$|a| = \min\{a, N - a\}, \quad \psi(g) = \sum_{j=1}^n |g_j|, \quad g \in \mathbb{Z}_N^n.$$

Let  $\mathcal{H}_\psi$  be the real cocycle Hilbert space obtained from the Gromov form

$$\langle \delta_g, \delta_h \rangle_\psi = \frac{1}{2} (\psi(g) + \psi(h) - \psi(g - h)).$$

For  $1 \leq r \leq m$  set  $\theta_r = 2\pi r/N$  and

$$q_r = - \sum_{a=0}^{N-1} |a| \cos(\theta_r a) = \frac{1 - (-1)^r \cos(\pi r/N)}{2 \sin^2(\pi r/N)} > 0.$$

For  $1 \leq j \leq n$ , define

$$c_{j,r} = - \frac{2}{\sqrt{N q_r}} \sum_{a=0}^{N-1} \cos(\theta_r a) \delta_{ae_j}, \quad s_{j,r} = \frac{2}{\sqrt{N q_r}} \sum_{a=0}^{N-1} \sin(\theta_r a) \delta_{ae_j}.$$

Then

$$\{c_{j,r}, s_{j,r} : 1 \leq j \leq n, 1 \leq r \leq m\}$$

is an orthonormal basis of  $\mathcal{H}_\psi$ .

Moreover, if  $\mathcal{H}_{\psi,j} = \text{span}\{c_{j,r}, s_{j,r} : 1 \leq r \leq m\}$ , then  $\mathcal{H}_\psi = \bigoplus_{j=1}^n \mathcal{H}_{\psi,j}$  is invariant under the cocycle action. The conditional expectations over the coordinate subgroups  $\mathbb{Z}_N^S$  form an admissible family of Fourier truncations. If

$$\partial_j \lambda(g) = \mathbf{1}_{\{g_j \neq 0\}} \lambda(g),$$

then  $\{\partial_j\}_{j=1}^n$  is a distinguished family of derivatives. Hence, for every  $p \geq 2$ , every  $1 \leq k \leq n$ , and every mean-zero  $f : \mathbb{Z}_N^n \rightarrow \mathbb{C}$ ,

$$\frac{1}{\binom{n}{k}} \sum_{\substack{S \subseteq [n] \\ |S|=k}} \left\| \sum_{g \in \mathbb{Z}_N^S} \widehat{f}(g) \exp\left(\frac{2\pi i}{N} \langle \cdot, g \rangle\right) \right\|_p^p \lesssim_p \frac{k}{n} \sum_{j=1}^n \|\partial_j f\|_p^p + \left(\frac{k}{n}\right)^{p/2} \|f\|_p^p.$$

## Proof

We first record the zero-sum form of the Gromov inner product. If  $x = \sum_g a_g \delta_g$  and  $y = \sum_h b_h \delta_h$  satisfy  $\sum_g a_g = \sum_h b_h = 0$ , then

$$\langle x, y \rangle_\psi = -\frac{1}{2} \sum_{g,h} a_g b_h \psi(g-h). \quad (1)$$

Indeed, the two terms containing  $\psi(g)$  and  $\psi(h)$  vanish after summing against the zero total coefficients.

Let  $\kappa(a) = |a|$  on  $\mathbb{Z}_N$ . For  $1 \leq r \leq m$  its nonzero Fourier coefficient is

$$\widehat{\kappa}(r) = \sum_{a=0}^{N-1} \kappa(a) e^{-i\theta_r a} = 2 \sum_{a=1}^m a \cos(\theta_r a) = -q_r. \quad (2)$$

The displayed closed form for  $q_r$  follows from

$$\sum_{a=1}^m a z^a = \frac{z(1 - (m+1)z^m + m z^{m+1})}{(1-z)^2}, \quad z^N = 1, \quad z \neq 1,$$

because  $z^m + z^{m+1} = 2(-1)^r \cos(\pi r/N)$  and  $z/(1-z)^2 = -1/(4 \sin^2(\pi r/N))$ .

Now fix one coordinate  $j$ . The real functions  $a \mapsto \cos(\theta_r a)$  and  $a \mapsto \sin(\theta_r a)$ ,  $1 \leq r \leq m$ , are the real nonzero Fourier modes of  $\mathbb{Z}_N$ . By (1) and (2),

$$\left\| \sum_a \cos(\theta_r a) \delta_{ae_j} \right\|_{\mathcal{H}_\psi}^2 = \left\| \sum_a \sin(\theta_r a) \delta_{ae_j} \right\|_{\mathcal{H}_\psi}^2 = \frac{N q_r}{4},$$

and different real modes are orthogonal. If  $j \neq j'$ , the coordinate supports are also orthogonal: by additivity of  $\psi$ , every mixed inner product is a sum of terms containing either the total coefficient of the  $j$ -mode or the total coefficient of the  $j'$ -mode, both of which are zero. The displayed normalization therefore gives an orthonormal family of  $2mn$  vectors. Since the source remark identifies  $\dim \mathcal{H}_\psi = 2mn$ , the family is an orthonormal basis.

The useful coefficient formulas are, for  $t = g_j \in \{0, \dots, N-1\}$ ,

$$\langle \delta_g, c_{j,r} \rangle_\psi = \sqrt{\frac{q_r}{N}} (1 - \cos(\theta_r t)), \quad \langle \delta_g, s_{j,r} \rangle_\psi = \sqrt{\frac{q_r}{N}} \sin(\theta_r t). \quad (3)$$

For instance, if  $C_r(a) = \cos(\theta_r a)$ , then

$$\left\langle \delta_g, \sum_a C_r(a) \delta_{ae_j} \right\rangle_\psi = \frac{1}{2} (\widehat{\kappa}(r) - \widehat{\kappa}(r) \cos(\theta_r t)) = -\frac{q_r}{2} (1 - \cos(\theta_r t)),$$

and the sign in the definition of  $c_{j,r}$  gives the first formula. The sine formula is the same convolution calculation.

The formulas (3) show in particular that every derivative in the  $j$ th coordinate vanishes on Fourier coefficients with  $g_j = 0$ . Consequently

$$\partial_{c_{j,r}} \circ \partial_j = \partial_{c_{j,r}}, \quad \partial_{s_{j,r}} \circ \partial_j = \partial_{s_{j,r}},$$

where  $\partial_u \lambda(g) = 2\pi i \langle \delta_g, u \rangle_\psi \lambda(g)$ . Thus the coordinate support derivatives form a distinguished family once the coordinate splitting is known to be invariant.

It remains only to note that invariance. For a zero-sum vector  $x = \sum_a x_a \delta_{ae_j}$ , the source cocycle action gives  $\alpha_g x = \sum_a x_a \delta_{g+ae_j}$  in the quotient. This is translation by  $g_j$  on the  $j$ th cyclic coordinate. Hence

$$\alpha_g c_{j,r} = \cos(\theta_r g_j) c_{j,r} - \sin(\theta_r g_j) s_{j,r}, \quad \alpha_g s_{j,r} = \sin(\theta_r g_j) c_{j,r} + \cos(\theta_r g_j) s_{j,r}.$$

So each  $\mathcal{H}_{\psi,j}$  is  $\alpha$ -invariant, and the coordinate subgroup conditional expectations over  $\mathbb{Z}_N^S$  are admissible exactly as in the even case treated in the source paper.

Applying Theorem B of [1] gives the stated inequality, up to the extra adjoint derivative terms present in the general theorem. Since the group is abelian and the multiplier  $\mathbf{1}_{\{g_j \neq 0\}}$  is symmetric under  $g \mapsto -g$ , we have  $\|\partial_j(f^*)\|_p = \|\partial_j f\|_p$ ; these terms are absorbed into the implicit constant depending only on  $p$ .

## Verification Notes

The script `code/verify_odd_cycle_onb.py` checks the finite Gram matrix and coefficient identities for  $N = 3, 5, 7, 9, 11$  and  $n = 1, 2, 3$  using the Gromov form directly. It is only a numerical sanity check; the proof above is symbolic.

Cheap-index search found no prior packet for arXiv:2209.05986. A web search for the exact odd-cardinal/orthonormal-basis phrases found the source preprint and its 2024 journal publication, but no separate explicit odd-cycle basis answer. The related arXiv:2209.05991 Part II problem has a prior no-promotion attempt in this run and is mathematically separate.

## Limitations and Human Review Recommendation

This packet should be reviewed as a compact completion of Remark 2.4, not as a solution to a broad open conjecture. The main review points are the sign and normalization in (2) and (3), and the use of Theorem B after verifying the distinguished derivative condition.

## References

- [1] Antonio Ismael Cano-Marmol, Jose M. Conde-Alonso, and Javier Parcet. Trigonometric chaos and  $X_p$  inequalities I – Balanced Fourier truncations over discrete groups. arXiv:2209.05986, 2022. Published in Analysis & PDE 17(7), 2561–2584, 2024.